GRAPH THEORY:
HOMEWORK 7
BOOK SECTION 3.1 PART III AND 4.1.

ROB TOMPKINS

1. Let M be a maximum matching and L a minimal edge cover in a graph with no isolated vertices. Probe the following statements below:

(a) M is a maximal matching if and only if $M$is contained in a minimum edge cover.
(b) L is a minimum edge cover if and only if L contains a maximal matching.

Proof. Write $n = |V(G)|$, let ν be the maximum matching size, and let ρ be the minimum edge cover size. We first show that $\rho = n - \nu$.

Every minimal edge cover F induces a disjoint union of stars, each with at least one edge. Indeed, an edge whose two endpoints both have degree at least two in $(V(G), F)$ could be deleted without uncovering a vertex. Thus every edge has a degree-one endpoint, which forces each component to be a star. If there are c such components, then $|F| = n - c$, and choosing one edge from each star gives a matching of size c. Consequently $|F| \geq n - \nu$. Applying this to a minimum edge cover gives $\rho \geq n - \nu$.

Conversely, start with a maximum matching M. Its unmatched vertices form an independent set, since an edge between two of them would enlarge the matching. Choose an incident edge for each unmatched vertex; this is possible because there are no isolated vertices. These edges are distinct and join unmatched vertices to matched vertices. Together with M they form an edge cover of size $\nu + (n - 2\nu) = n - \nu$. Hence $\rho = n - \nu$.

(a) Under the opening assumption that M is a *maximum* matching, both assertions are true: M is maximal, and the preceding construction extends it to a minimum edge cover. Thus the stated equivalence holds under that assumption. If the intended statement instead quantifies over arbitrary matchings, "maximal" must be replaced by "maximum." For example, on the path $1, 2, 3, 4$, the matching $\{23\}$ is maximal, but the unique minimum edge cover is $\{12, 34\}$.

For completeness, a matching contained in a minimum edge cover need not itself be maximum (it could be empty). The usual unrestricted equivalence requires the matching to be *maximal*: a maximal matching is maximum if and only if it is contained in a minimum edge cover. To see the reverse direction, a minimum edge cover has exactly ν star components. A maximal matching contained in it must choose one edge from every star, and therefore has size ν.

(b) As written, this statement is false. On the path $1, 2, 3, 4, 5, 6$, let

$$L = \{12, 23, 45, 56\}.$$

This is a minimal edge cover, since deleting any of its edges uncovers a leaf of one of its two star components. It contains the maximal matching $\{12, 45\}$: the only unmatched vertices are 3 and 6, which are not adjacent. However, L is not minimum, since $\{12, 34, 56\}$ is an edge cover with only three edges.

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The corrected assertion replaces "maximal" by "maximum." If L is minimum, its star decomposition has $n-|L| = \nu$ components, so selecting one edge from each gives a maximum matching contained in L. Conversely, if the minimal edge cover L contains a maximum matching, each of its c star components contributes at most one matching edge. Hence $\nu \le c \le \nu$, so $|L| = n - c = n - \nu = \rho$, and L is minimum. □

2. Let G be a simple graph in which the sum of degrees of any k vertices is less than $n-k$. Prove that every maximal independent set in G has more than k vertices.

Proof. Let I be a maximal independent set, and suppose $|I| \le k$. Choose a set S of k vertices containing I. (As usual the hypothesis is understood with $0 \le k \le n$.) Maximality implies that every vertex outside I, and hence every vertex outside S, has a neighbor in I. Thus at least $n-k$ edges join I to $V(G) \setminus S$, giving

$$n - k \le \sum_{v \in I} d(v) \le \sum_{v \in S} d(v) < n - k,$$

a contradiction. Therefore $|I| > k$. □

3. An edge e of a graph G is α-critical is $\alpha(G-e) > \alpha(G)$. Suppose that xy and xz are α-critical edges in G. Prove that G has an induced subgraph that is an odd cycle containing xy and xz. (Hint: Let Y, Z be maximum independent sets in $G-xy$ and $G-xz$ respectively. Let $H = G[Y \Delta Z]$. Prove that every component of H has the same number of vertices from Y to Z. Use this to prove y and z belong to the same component of H.)

Proof. Put $a = \alpha(G)$. Deleting one edge increases the independence number by at most one: removing one endpoint of that edge from an independent set in the edge-deleted graph leaves an independent set in G. Therefore the sets in the hint satisfy $|Y| = |Z| = a + 1$. Moreover $x, y \in Y$ and $x, z \in Z$; otherwise one of these sets would already be independent in G. Since xz is present in $G - xy$ and xy is present in $G - xz$, we have $z \notin Y$ and $y \notin Z$.

Set $A = Y \setminus Z$, $B = Z \setminus Y$, and $H = G[A \cup B]$. The sets $Y \setminus \{x\}$ and $Z \setminus \{x\}$ are independent in G and have size a, so both are maximum. In particular H is bipartite with parts A, B.

Let C be any component of H. Exchanging $A \cap V(C)$ for $B \cap V(C)$ in $Y \setminus \{x\}$ preserves independence: there are no edges to another component of H, and vertices of $(Y \cap Z) \setminus \{x\}$ have no neighbors in either part. Maximality of the cardinality a implies $|B \cap V(C)| \le |A \cap V(C)|$. Exchanging in the other direction using $Z \setminus \{x\}$ gives the reverse inequality. Hence each component has equally many vertices in A and B.

Suppose y and z lie in different components, and let C contain y. Then

$$Y' = (Y \setminus (A \cap V(C))) \cup (B \cap V(C))$$

has size $a + 1$. The same exchange argument shows that its vertices other than x are independent. The only neighbor of x in Y is y, which was removed, and the only neighbor of x in Z is z, which was not added. Thus Y' is independent in G, a contradiction. Therefore y and z belong to the same component.

Choose a shortest y–z path P in H. Because $y \in A$ and $z \in B$, its length is odd. A shortest path has no chord, and x has no neighbors in $V(H) \setminus \{y, z\}$. Consequently P together with the edges xy, xz is an induced cycle of odd length, containing both specified edges. □

4. Let G be a connected graph with at least three vertices. Form G' and G by adding an edge with endpoints x, y whenever $d_G(x, y) = 2$. Prove that G' is 2-connected.

Proof. Interpret the construction as adding the specified edges to G. Fix any vertex v. For any two distinct vertices $a, b \neq v$, choose a simple a–b path in G. If it avoids v, it remains a path in $G' - v$. Otherwise let u, v, w be the consecutive vertices where it passes through v. The distinct vertices u, w are either adjacent in G or have distance two in G, so $uw \in E(G')$. Replacing u, v, w by u, w gives an a–b path in $G' - v$. Thus $G' - v$ is connected for every v. Since G' is connected and has at least three vertices, it is 2-connected. □

5. Prove that $\kappa(G) = \delta(G)$ if G is simple and $\delta(G) \geq n(G) - 2$. Prove that this is the best possible for each $n \geq 4$ by constructing a simple n-vertex graph with minimum degree $n - 3$ and connectivity less than $n - 3$.

Proof. Write $n = n(G)$, using the standard convention $\kappa(K_n) = n - 1$. If G is complete, the desired equality is immediate. Otherwise $\delta(G) = n - 2$, and each vertex has at most one nonneighbor among the other vertices.

Suppose S is a vertex set with $|S| < n - 2$. At least three vertices remain. If $G - S$ were disconnected, choose vertices u, v in two distinct components, and a third remaining vertex w. If w is in the component of u, then v has the two nonneighbors u, w; otherwise u has the two nonneighbors v, w. Both are impossible. Hence deleting fewer than $n-2$ vertices cannot disconnect G. On the other hand, since G is not complete, choose nonadjacent vertices p, q. Deleting the other $n - 2$ vertices leaves two isolated vertices. Thus $\kappa(G) = n - 2 = \delta(G)$.

For sharpness, for each $n \geq 4$ take two disjoint copies of K_2 and a clique S on $n - 4$ vertices, and join every vertex of S to all four vertices in the two copies of K_2. Each of those four vertices has degree $1 + (n - 4) = n - 3$, while each vertex in S has degree $n - 1$. Therefore $\delta(G) = n - 3$. Deleting S leaves $K_2 \dot{\cup} K_2$, so

$$\kappa(G) \leq n - 4 < n - 3 = \delta(G).$$

When $n = 4$, S is empty and the graph is already disconnected, with connectivity zero. Thus the construction covers every $n \geq 4$. □